

Hawkeye EQMS · Complaint Management · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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Tenant under test	Novex Pharma Inc. · 69e64e7869b2ba745d40bb89
Live frontend	https://hawkeye-frontend-dev-chi.vercel.app
Live backend	https://hawkeye-backend-dev.vercel.app

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1. Module overview

Module	Complaint Management
Pharma purpose	Capture, investigate, link to CAPA, and close customer complaints. Determine MDR (Medical Device Report) reportability. Track regulatory reporting timelines.
Compliance basis	ISO 9001:2015 §10.2 · 21 CFR 211.198 (drug complaint files) · 21 CFR 820.198 (medical device complaint files) · EU GVP
Hawkeye routes	/complaint-manager (frontend) · /api/complaints (backend)
Hawkeye model file	backend/src/models/ComplaintModel.js
Tenant module flag	modules.COMPLAINT_MANAGEMENT

Complaints follow OPEN → UNDER_INVESTIGATION → PENDING_CAPA → CAPA_IN_PROGRESS → PENDING_CLOSURE → CLOSED (or CANCELLED). isMedicalDeviceReport flag triggers the MDR reportability assessment; requiresRegulatoryReporting gates the regulatory submission timeline. linkedCAPAIds bridges to the CAPA-v2 workspace.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped.

MET = parity, **PARTIAL** = ships but with caveats, **GAP** = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
Complaint types (PRODUCT_QUALITY / LABELING / PACKAGING / DELIVERY / SERVICE / SAFETY / REGULATORY / OTHER) 21 CFR 820.198	complaintType enum · 8 values.	MET	
Severity (CRITICAL / MAJOR / MINOR / INFORMATIONAL) —	severity enum.	MET	
Source tagging (CUSTOMER / PATIENT / REGULATOR / DISTRIBUTOR / INTERNAL / FIELD_REPORT) —	source enum · 7 values.	MET	
MDR reportability flag 21 CFR 803	isMedicalDeviceReport boolean. cross-module triggers regulatory assessment.	MET	
Regulatory reporting flag + 5-day / 30-day timer 21 CFR 803.10	Auto-computed on save: isMedicalDeviceReport OR severity=CRITICAL OR complaintType=SAFETY → requiresRegulatoryReporting=true + regulatoryDeadlineDays + mdrDueDate (5d MDR / 15d critical / 30d otherwise).	MET	
Investigation summary + root cause + assignee 21 CFR 820.198(d)	POST /:id/investigate captures investigationSummary + rootCause + assignedTo.	MET	
Linked CAPA(s) 21 CFR 820.198(c)	linkedCAPAIds[] on the model. Cross-module endpoint creates CAPA-v2 intake with sourceType=COMPLAINT.	MET	
Closure with corrective + preventive action notes 21 CFR 820.198(e)	POST /:id/close captures closureNotes + correctiveAction + preventiveAction.	MET	
Auto-numbered complaint (COM-YYYY-NNNN) GMP traceability	complaintNumber pre-save generator.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
James Thompson Head of QA · complaint owner	<code>qa.head@novex-pharma.demo</code>	Logs complaints from intake (customer service), investigates, links CAPA, closes.	OPEN UNDER_INVESTIGATION PENDING_CAPA PENDING_CLOSURE CLOSED
Marcus Brown Regulatory Affairs · MDR reviewer	<code>regulatory@novex-pharma.demo</code>	Assesses MDR reportability for safety / patient complaints; files MDR with FDA.	OPEN UNDER_INVESTIGATION

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Audit Lead Maria auditor · linked-CAPA owner	audit.lead@novex-pharma.demo	Owens the CAPA spawned from the complaint.	CAPA_IN_PROGRESS

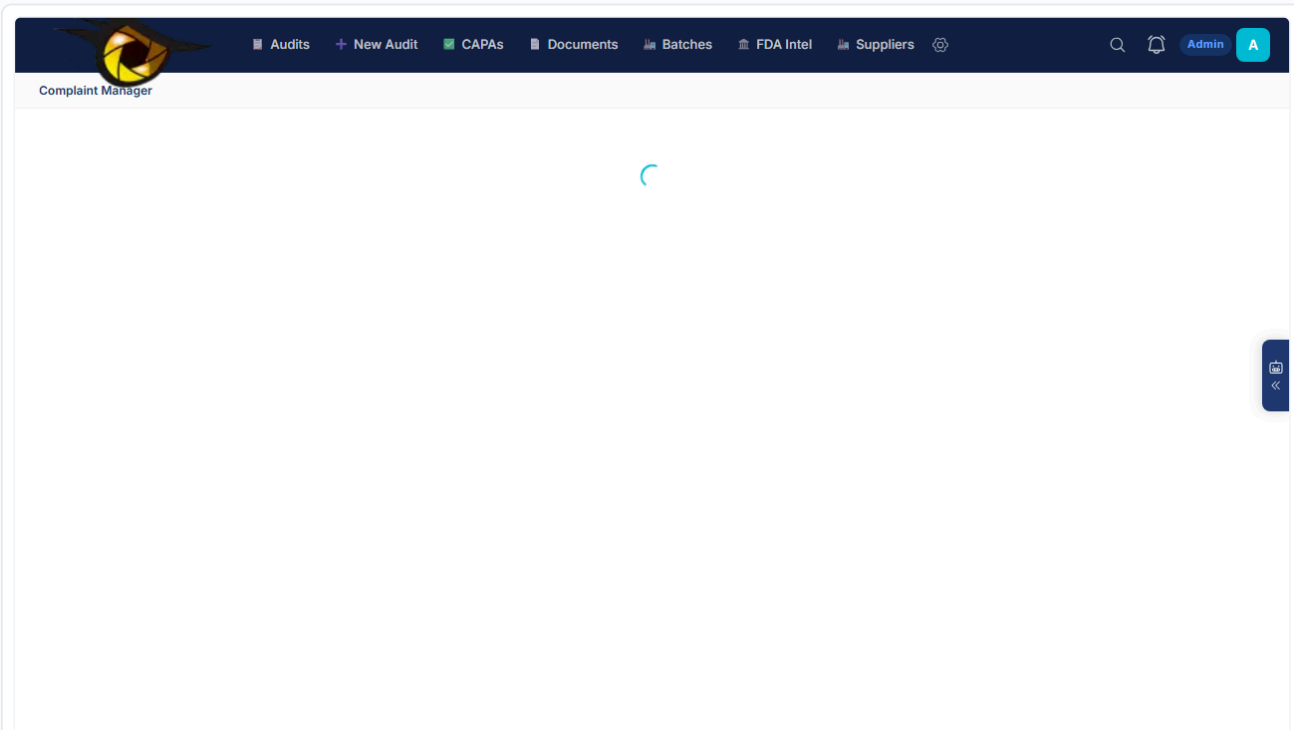
4. Feature catalogue (click-by-click)

4.1 Complaint register

What it does	List with severity / status / type filters + MDR + regulatory-reporting indicators.
Menu / route	/complaint-manager
Available to roles	any tenant viewer
Backend API	GET /api/complaints

CLICK-BY-CLICK

- NAVIGATE** Click 'Complaints' in the top nav
→ Page renders
- WAIT** Spinner clears
→ Rows visible (empty for fresh tenant)



state-screens/complaint-list.png

4.2 + Log Complaint dialog

What it does	Capture a new complaint at intake.
Menu / route	/complaint-manager · top-right '+ Log Complaint' button
Available to roles	any tenant user

CLICK-BY-CLICK

1. **CLICK** Click '+ Log Complaint'
→ Dialog opens
2. **TYPE** Fill title (1 sentence)
→ required
3. **TYPE** Fill description (full narrative)
→ required
4. **CLICK** Pick complaintType
→ 8 enums
5. **CLICK** Pick severity
→ 4 enums
6. **CLICK** Pick source
→ 7 enums
7. **CLICK** Tick isMedicalDeviceReport if device-related
→ drives MDR assessment
8. **CLICK** Click 'Save'
→ Row appears with status=OPEN; complaintNumber=COM-YYYY-NNNN

FIELDS CAPTURED

Field	Required	Validation / values	Notes
title	Yes	string	
description	Yes	multi-line	
complaintType	Yes	PRODUCT_QUALITY LABELING PACKAGING DELIVERY SERVICE SAFETY REGULATORY OTHER	
severity	Yes	CRITICAL MAJOR MINOR INFORMATIONAL	
source	Yes	CUSTOMER PATIENT REGULATOR DISTRIBUTOR INTERNAL FIELD_REPORT OTHER	
isMedicalDeviceReport	No	boolean	default false

4.3 Investigate (OPEN → UNDER_INVESTIGATION)

What it does	Capture investigation summary, root cause, assignee.
Menu / route	Complaint row · 'Investigate' button
Available to roles	any tenant user
Backend API	POST /api/complaints/:id/investigate

CLICK-BY-CLICK

1. **CLICK** Click 'Investigate'
→ Drawer opens
2. **TYPE** Fill investigationSummary
→ required
3. **TYPE** Fill rootCause
→ required
4. **CLICK** Pick assignedTo from user dropdown

→ required

5. **CLICK** Click 'Submit'
 - Status=UNDER_INVESTIGATION; investigationCompletedAt set

4.4 Link CAPA

What it does	Bridge to the CAPA-v2 workspace by creating an intake with sourceType=COMPLAINT.
Menu / route	Complaint row · 'Create CAPA' button
Available to roles	any tenant user
Backend API	POST /api/capa-v2/intakes (sourceType=COMPLAINT) · then PUT /api/complaints/:id (status=PENDING_CAPA, linkedCAPAIds += new id)

CLICK-BY-CLICK

1. **CLICK** Click 'Create CAPA' on the row
 - Drawer with CAPA intake fields
2. **TYPE** Fill issueStatement + issueDescription + severity
 - required
3. **CLICK** Click 'Create + link'
 - CAPA-v2 intake row created; linkedCAPAIds += new id; complaint status=PENDING_CAPA

4.5 Close

What it does	Close the complaint after investigation + (linked CAPAs) closed.
Menu / route	Complaint row · 'Close' button
Available to roles	any tenant user
Backend API	POST /api/complaints/:id/close

CLICK-BY-CLICK

1. **CLICK** Click 'Close'
 - Drawer with closureNotes + correctiveAction + preventiveAction
2. **TYPE** Fill closureNotes (required)
 - required
3. **TYPE** (Optional) correctiveAction + preventiveAction
4. **CLICK** Click 'Submit'
 - Status=CLOSED; closedAt + closedBy set

4.6 Cross-module · Regulatory assessment

What it does	Compute regulatory reporting required + flags + recommended deadline.
Menu / route	Complaint detail · 'Assess regulatory reporting' button
Available to roles	any tenant user
Backend API	POST /api/quality/regulatory-assessment

CLICK-BY-CLICK

1. **API** POST body = the full complaint object
 - Returns { requiresReporting, flags[], recommendedDeadlineDays, regulatoryBodies[] }

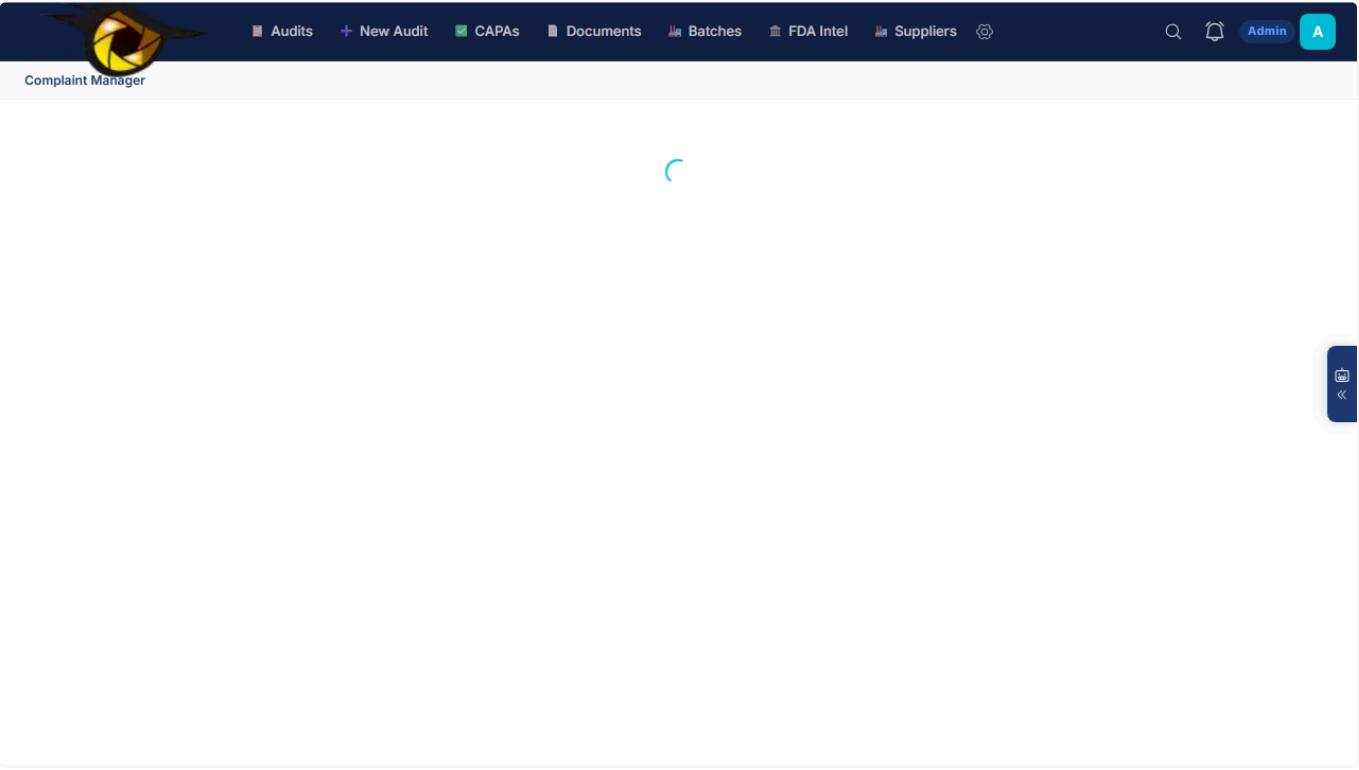
5. Full lifecycle walkthrough — one transaction, end-to-end

One complaint walked OPEN → UNDER_INVESTIGATION → CAPA-linked → CLOSED.

#	Persona	State transition	Action	API	Observed	Result
1	James Head of QA	– → OPEN	+ Log Complaint → 'Chipped tablets at customer' / PRODUCT_QUALITY / MAJOR / CUSTOMER → Save	POST /api/complaints	complaintNumber=COM-NNNN · status=OPEN	PASS

Expected DB state after this step:

```
complaints { _id, complaintNumber: 'COM-NNNN', title, description, complaintType: 'PRODUCT_QUALITY', severity: 'MAJOR', source: 'CUSTOMER', status: 'OPEN', reportedBy: james._id }
```



state-screens/complaint-list.png

2	James Head of QA	OPEN → UNDER_INVESTIGATION	Click 'Investigate' → fill summary + rootCause + assignedTo=james → Submit	POST /api/complaints/:id/investigate	Status flips · investigationCompletedAt set	PASS
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Expected DB state after this step:

```
complaints { status: 'UNDER_INVESTIGATION', investigationSummary, rootCause, assignedTo, investigationCompletedAt }
```

3	James Head of QA	UNDER_INVESTIGATION → PENDING_CAPA	Click 'Create CAPA' → CAPA-v2 intake created with sourceType=COMPLAINT → linked	POST /api/capa-v2/intakes · PUT /api/complaints/:id (linkedCAPAIds, status=PENDING_CAPA)	linkedCAPAIds += [intakeId]; status=PENDING_CAPA	PASS
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#	Persona	State transition	Action	API	Observed	Result
Expected DB state after this step:						
<pre>capa-v2-intakes { sourceType: 'COMPLAINT', sourceRecordId: complaintId } complaints { status: 'PENDING_CAPA', linkedCAPAIds: [intakeId] }</pre>						
4	James Head of QA	PENDING_CAPA → PENDING_CLOSURE	(After CAPA close) PUT status=PENDING_CLOSURE	PUT /api/complaints/:id	Status flips	PASS
5	James Head of QA	PENDING_CLOSURE → CLOSED	Click 'Close' → closureNotes + correctiveAction + preventiveAction → Submit	POST /api/complaints/:id/close	Status=CLOSED · closedBy + closedAt set	PASS
Expected DB state after this step:						
<pre>complaints { status: 'CLOSED', closureNotes, correctiveAction, preventiveAction, closedBy, closedAt }</pre>						

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
(roadmap) Complaint triage agent (Wave 3)	OPEN	(future)	(future)	Pattern-match complaint text against historical CAPAs + FDA MedWatch to pre-suggest severity, MDR reportability, and linked CAPAs	Free Gemini + vector retrieval
Cross-module Regulatory Assessment	OPEN	POST /api/quality/regulatory-assessment	(API; UI on roadmap)	Returns requiresReporting + flags + recommendedDeadlineDays + regulatoryBodies	rule engine (no LLM)

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
OPEN	21 CFR 820.198(a) 21 CFR 211.198(a)	complaintNumber + title + complaintType + severity + source + reportedBy	complaints
UNDER_INVESTIGATION	21 CFR 820.198(d)	investigationSummary + rootCause + assignedTo + investigationCompletedAt	complaints
PENDING_CAPA	21 CFR 820.198(c)	linkedCAPAIds[]	complaints + capa-v2 (linked)
CLOSED	21 CFR 820.198(e) 21 CFR 211.198(b)	closureNotes + correctiveAction + preventiveAction + closedBy + closedAt	complaints

8. Test results matrix

Suite	Scope	Result	Evidence
eqms-lifecycle.spec.ts · complaint	OPEN → UNDER_INVESTIGATION → PENDING_CLOSURE → CLOSED	PASS	4/4 PASS · eqms-test-results-v2.pdf
eqms-cross-module.spec.ts · F3	Complaint → CAPA-v2 intake (sourceType=COMPLAINT) → linked → close	PASS	5/5 PASS · F3

9. Known gaps + roadmap

- **AI complaint triage agent** — Pre-suggest severity, MDR flag, similar past CAPAs. **HIGH**
- **MDR / FAR deadline countdown UI** — Visible counter on the row when requiresRegulatoryReporting=true. Email reminder at 80% elapsed. **HIGH**
- **FDA MedWatch eMDR submission integration** — Programmatic 3500A submission API. **MEDIUM**
- **Customer notification template** — Closure notes auto-formatted into a customer-facing letter. **LOW**